

Swirl Gated Curvature Correction

Summary of research approach — RANS curvature/swirl correction for hydrocyclone CFD.

1. Core idea

Standard two-equation RANS models (k - ϵ , k - ω SST, realizable k - ϵ) are known to over-predict eddy viscosity in curved/swirling flows, over-diffusing the vortex structures that matter most in the target application (hydrocyclone separation). Rather than applying a curvature correction uniformly everywhere (the older BCS1 / BCS2 approach), the current generation (BCS3 / BCS3_kande) adds a **local swirl gate**:

```
F_swirl = clamp( |uθ| / √(uθ2 + uz2 + ε), 0, 1 )
frEff = 1 + F_swirl · (fr - 1) [swirlCorrection = true, "CCS"]
frEff = fr [swirlCorrection = false, "CC"]
```

`fr` (the Richardson-number-based curvature factor) only takes full effect where the flow is genuinely swirl-dominated locally; it fades toward inert (`frEff=1`) elsewhere. This local gating — rather than applying the correction everywhere a streamline happens to be curved — is the actual novel mechanism of this project.

2. Structure of the study

- **Three correction levels** tested consistently: base (no correction) → CC (curvature only, gate off) → CCS (curvature + swirl gate).
- **Applied across multiple base models**, not just one: `k0omegaSST` , `realizableKE` , `kEpsilon` , `k0omega` , `RNGkEpsilon` all have `_BCS3` / `_BCS3_kande` variants — testing whether the correction generalizes across model families rather than being tuned to a single one.

Two validation targets, deliberately separated

Case	Role	Details
IWA mini-hydrocyclone (Liu et al. 2022)	Actual target application	Multiphase solver (<code>incompressibleDriftFlux</code>), validated against PIV axial-velocity profiles at 4 heights, across coarse/medium/fine mesh resolution.
Dellenback et al. swirling pipe-past-expansion (S=0.6)	Clean literature benchmark	Decoupled from hydrocyclone-specific complexity (multiphase, complex geometry); real digitised experimental inlet swirl profile mapped via <code>timeVaryingMappedFixedValue</code> . Isolates whether the swirl-gate mechanism itself works before trusting it in the harder geometry.

Primary metric: R^2 against experimental data, computed per model $\times$ correction level $\times$ downstream station.

3. What the data has shown so far

CC vs CCS differences are small in both validation cases — in the pipe case, plain `realizableKE` baseline slightly *out-scored* its own CC and CCS variants on average R^2 ; `kOmegaSST-CCS` only marginally beat plain `kOmegaSST`. Not yet a clean "correction helps" result.

All models — corrected or not — share the same failure mode far downstream ($z/D=3$ axial velocity, negative R^2 universally), suggesting a shared limitation (likely the recirculation/reattachment region) rather than something curvature correction specifically addresses.

`RSM` (the natural gold-standard anisotropic-turbulence comparison) hasn't yet produced converged comparable data, so the picture is still incomplete.

4. Methodological pitfalls caught along the way

Worth remembering as lessons, not just history.

- A "CC" case was once silently misconfigured identically to "CCS" (`swirlCorrection true` in both) — would have invalidated the whole comparison if unnoticed.
- An aggressive Courant-ramp solver schedule (`maxCo` up to 5) that works fine for the hydrocyclone cases turned out to badly over-dissipate swirl in the pipe case specifically — large timesteps under-resolve rotational dynamics. A "one size fits all" solver-settings template does not transfer safely between a separation-flow case and a rotation-dominated validation case.
- Meshes, initial conditions, and custom-library paths are validated case-by-case (`checkMesh`, BC consistency, library linkage) before committing cluster time — a deliberate cheap-local-check-before-expensive-cluster-run workflow throughout.

Summary of the "Swirl Gated Curvature Correction" research approach, IWA mini-hydrocyclone / Dellenback swirling-pipe validation.

Source: OpenFOAM-12 custom `libmycompressibleMomentumTransportModels` library.